

An Introduction to Higher Algebraic K-Theory

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Chapter 1

Projective Modules

The subject of K-Theory concerns with studying vector bundles on spaces. Algebraic K-Theory studies the algebraic analogue of the same, projective modules over associative rings.

1.1 Projective Modules

1.1.1 Idempotence Criteria for Projective Modules

1.1.2 Constructing Projective Modules

- Patching along localizations
- Milnor Patching

1.1.3 Classification of Projective Modules

- Projective modules $\leftrightarrow$ vector bundles -

Chapter 2

The Grothendieck Group K_0

In this chapter, we will develop the notion of the Grothendieck Group K_0 across a variety of contexts.

2.1 K_0 of a ring

Let R be a (unital) ring. The interesting objects of study will be finitely generated projective modules over R . We denote by $\mathbf{P}(R)$ the set of isomorphism classes of finitely generated projective modules over R . Note the $\mathbf{P}(R)$ derives a monoid structure through $\oplus$ because direct sums yield an associative operation that preserves isomorphisms. This monoid is also abelian due to the isomorphisms $M \oplus N \cong N \oplus M$ for any R -modules.

The idea is to consider the group naturally derived by implementing inverses for all elements in this monoid.

2.1.1 Group Completion of a Monoid

Given an (abelian) monoid $(M, +)$, we can derive an (abelian) group associated to M , which we shall call $\tilde{M}$, via the following procedure:

Define an equivalence relation on $M \times M$ by

$$(a, b) \sim (c, d) \Leftrightarrow a + d + e = b + c + e \text{ for some } e \text{ in } M.$$

This indeed gives us an equivalence relation. Let $\tilde{M}$ be the set of equivalence classes of $M \times M$ under this relation. This set admits a natural (abelian) group operation defined by

$$[(a, b)] + [(c, d)] := [(a + c, b + d)]$$

This is a well-defined, abelian and associative operation on $\tilde{M}$ with the

identity element $0 := [(0, 0)]$ represented by any (l, l) . Each element $[(a, b)]$ admits an inverse given by $[(b, a)]$ as $[(a, b)] + [(b, a)] = [(a + b, b + a)] = 0$.

A very familiar example of this construction is the abelian group $\mathbb{Z}$, which is the group completion of the abelian monoid $\mathbb{N}$ under $+$.

This construction is natural in M and follows a suitable universal property which we shall state.

Proposition 1 (Universal property of Group Completion). *We have a monoid homomorphism $i : M \rightarrow \tilde{M}$ such that for any (abelian) group G , and a monoid homomorphism $\varphi : M \rightarrow G$, we have that φ must factor uniquely through i . That is, there is a unique map $\tilde{\varphi} : \tilde{M} \rightarrow G$ such that the following diagram commutes:*

$$\begin{array}{ccc} M & \xrightarrow{i} & \tilde{M} \\ & \searrow \varphi & \downarrow \tilde{\varphi} \\ & & G \end{array}$$

Proof. The monoid homomorphism $i : M \rightarrow \tilde{M}$ is simply given by mapping a to $[(a, 0)]$. This is indeed a homomorphism since $[(a, 0)] + [(b, 0)] = [(a + b, 0)]$. Now suppose we are given $\varphi : M \rightarrow G$. Then $\tilde{\varphi} : \tilde{M} \rightarrow G$ is given by $[(a, b)] \mapsto \varphi(a) - \varphi(b)$, making the diagram commute. This map is unique since for any $a \in M$, we have $\tilde{\varphi}([(a, 0)]) = \varphi(a)$, and consequently $\tilde{\varphi}([(0, a)]) = -\tilde{\varphi}([(a, 0)]) = -\varphi(a)$, which altogether determines the map $\tilde{\varphi}$. $\square$

Furthermore, any abelian group satisfying the given universal property is isomorphic to $\tilde{M}$ via a unique isomorphism.

There is an alternative (but equivalent) characterization of $\tilde{M}$ via the quotienting the free group on M by the monoidal relations.

Proposition 2. *Let $K(M)$ be the abelian group given by taking the free abelian group $F(M)$ on the set M and quotienting by all relations of the form $[a] + [b] - [a + b]$ where $a, b \in M$ and $[x]$ represents the element in $F(M)$ corresponding to the element $x \in M$. Then $K(M) \cong \tilde{M}$.*

Proof. We will show that $K(M)$ satisfies the universal property of $\tilde{M}$. Let $F(M)$ represent the free abelian group on the set M . The map $i : M \rightarrow K(M)$ on x is given by taking the equivalence class of the element x . Suppose $\varphi : M \rightarrow H$ is a monoid homomorphism to an abelian group H . Then define $\tilde{\varphi}([a])$ as $\varphi(a)$. Suppose $[a] = [b]$, then $[a] + [-b] = [0]$ so $a - b = 0$ and consequently $\varphi(a) - \varphi(b) = 0$. Thus the map $\tilde{\varphi}$ is well-defined and clearly a group homomorphism. It is also unique since $\tilde{\varphi}$ is specified uniquely on the basis of $F(M)$. $\square$

Example 2.1.1. Let R be a (unital) ring satisfying the Invariant Basis Property (IBP). That is, $R^m \cong R^n$ implies $m = n$. Then the monoid of isomorphism classes of free R -modules with the operation $\oplus$ is isomorphic to $\mathbb{N}$, the isomorphism given by the map $R^n \mapsto n$. Consequently, its group completion is isomorphic to the abelian group $\mathbb{Z}$.

Note that all fields (and division rings) F satisfy IBP since dimension of an F -module is independent of the choice of basis. Further, any ring R with a map $R \rightarrow F$, a field or a division ring F satisfies IBP. In particular, commutative rings all satisfy IBP.

Example 2.1.2. Let X be a topological space. Then the set $[X, \mathbb{N}]$ of continuous maps $X \rightarrow \mathbb{N}$ forms an abelian monoid under pointwise addition of maps. Then the group completion of this monoid is isomorphic to the abelian group of continuous maps $X \rightarrow \mathbb{Z}$ denoted $[X, \mathbb{Z}]$. If X is quasicompact, this is in fact a free abelian group. Notably, $[X, \mathbb{N}]$ also admits a commutative semiring structure via pointwise multiplication in addition to the monoid operation. This makes $[X, \mathbb{Z}]$ into a commutative ring.

2.1.2 Defining $K_0(R)$

Now we can define the group $K_0(R)$ as the group completion of the monoid $\mathbf{P}(R)$ consisting of finitely generated projective R -modules with the monoid operation given by $\oplus$.

Example 2.1.3. Let F be a field or a division ring. Then all finitely generated projective F -modules are free. This gives us $K_0(F) \cong \mathbb{Z}$. More generally, for any local ring R , all finitely generated projective R -modules are free. This gives us $K_0(R) \cong \mathbb{Z}$.

2.1.3 Rank and H_0

Recall the notion of rank of a projective module. Let R be a commutative ring and P be a projective R -module. For any prime ideal $\mathfrak{p}$ of R , we can look at the local ring $R_{\mathfrak{p}}$. Its quotient by the maximal ideal $k(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$ is a field and $P \otimes k(\mathfrak{p})$ becomes a projective $k(\mathfrak{p})$ -module, which must be free since $k(\mathfrak{p})$ is a field. Denote its rank by $\text{rank}_P(\mathfrak{p})$. We call this the local rank of P at $\mathfrak{p}$. If P has the same local rank for all prime ideals of R , we say P is of constant rank.

In general, we define $H_0(R)$ to be the ring $[Spec(R), \mathbb{Z}]$, the ring of all continuous maps from $Spec(R)$ to the discrete space $\mathbb{Z}$. Since $Spec(R)$ is quasicompact, it has finitely many connected components. Note that

$H_0(R)$ is a free abelian group with a basis given by maps of the We obtain a functrank : $\text{Spec}(R) \rightarrow$

Note there is an inclusion $[\text{Spec}(R), \mathbb{N}] \hookrightarrow \mathbf{P}(R)$ of monoids, which yields an inclusion of their corresponding group completions $H_0(R) \hookrightarrow K_0(R)$. This inclusion splits the rank map, and we get

$$K_0(R) \cong H_0(R) \oplus \tilde{K}_0(R)$$

where $\tilde{K}_0(R)$ is the kernel of the rank map.

Example 2.1.4.

2.1.4 Morita Equivalence and Invariance

Suppose we have (unital) rings R and S such that we have an equivalence of the respective categories of right modules:

$$\mathbf{mod}\text{-}R \simeq \mathbf{mod}\text{-}S$$

Then we say that R and S are **Morita equivalent**.

Chapter 3

Groups K_1 and K_2 of a ring

Chapter 4

Constructions of Higher K-Theory: Quillen's $+$ and Q

4.1 Quillen's $+=Q$ Theorem

Chapter 5

Some Miscellaneous Computations